

Corollary: e idempotent of R , the followings are equivalent.

- (1) e is primitive
- (2) eR is indecomposable as a left R -module
- (3) e is the unique idempotent in eRe

Recall. e idempotent of R , if $e = e_1 + e_2$ is an idempotent decomposition, then $e_1, e_2 \in eRe$, as $ee_1e = e_1$, $ee_2e = e_2$

Theorem: $e, f \in R$, idempotents, $V: R$ -module

(1) $\text{Hom}_R(eR, V) \cong eV$, in particular, $\text{Hom}_R(eR, Rf) \cong eRf$

(2) $\text{End}_R(eR) \cong (eRe)^{\text{op}}$ as rings

$\uparrow$
(as abelian gp)

proof: (1) Let $\varphi \in \text{Hom}_R(eR, V)$

then $\varphi(e) = \varphi(ee) = e\varphi(e) \in eV$

Consider $g: \text{Hom}_R(eR, V) \rightarrow eV$
 $\varphi \mapsto \varphi(e)$

$\varphi(ae) = a\varphi(e)$, hence φ is uniquely determined by $\varphi(e)$

Hence g is injective.

On the other hand, let $v \in eV$

let $\varphi: eR \rightarrow V$ s.t. $xe \mapsto xv$

then $g(\varphi) = \varphi(e) = ev = v$

Therefore g is bijective.

Only need to check g is a homomorphism.

(2). Let $\sigma, \tau \in \text{End}_R(R_e)$

$$\sigma(\tau(e)) = \sigma(\tau(e)e) = \tau(e)\sigma(e) \quad \text{hence by "}"$$

$$\text{we have } \text{End}_R(R_e) \cong (eRe)^{\text{op}}$$

Theorem: Let $e, f \in R$ idempotents, the followings are equivalent:

(1). $R_e \cong R_f$ as left module

(2). $eR \cong fR$ as right module

(3). There exists $b \in fRe$, $a \in eRf$ with $ab = e$ $ba = f$

proof: (1) $\Leftrightarrow$ (2) obvious

$$(1) \Rightarrow (3)$$

Let $\varphi: R_e \rightarrow R_f$ be an isomorphism

$$\text{Let } a = \varphi(e) \quad b = \varphi^{-1}(f)$$

$$\text{Then } a = \varphi(ee) = e\varphi(e) \in eRf$$

Similarly, $b \in fRe$

$$f = \varphi(\varphi^{-1}(f)) = \varphi(b) = \varphi(be) = b\varphi(e) = ba$$

$$e = \varphi^{-1}(\varphi(e)) = \varphi^{-1}(a) = \varphi^{-1}(af) = a\varphi^{-1}(f) = ab$$

$$(3) \Rightarrow (1)$$

$$eaf = eaba = ea = a$$

$$fbe = fba = fb = b$$

$$\text{hence: } \varphi_a: R_e \rightarrow R_f \quad \text{s.t.} \quad xe \mapsto xa$$

$$\varphi_b: R_f \rightarrow R_e \quad \text{s.t.} \quad xf \mapsto xb$$

are two R -homomorphisms and $\varphi_a \circ \varphi_b = \text{id}_{R_f}$ $\varphi_b \circ \varphi_a = \text{id}_{R_e}$

Jacobson Radical of a Ring

Definition: The intersection of all maximal left ideals of R is called the Jacobson Radical of R , denoted by $J(R)$

theorem: The following holds for the radical of a ring R

(1) $J(R)$ is the intersection of all the annihilator ideals of simple left R -modules. In particular, $J(R)$ is a 2-sided ideal of R

(2) $a \in J(R) \Leftrightarrow 1 - xa$ has left inverse for all $x \in R$

$$\text{i.e. } \exists b \in R \text{ s.t. } b(1 - xa) = 1$$

(3) $J(R)$ is the largest one among the 2-sided ideal of R satisfying the following condition

$$a \in J \Rightarrow 1 - a \text{ is a unit in } R$$

(4) $J(R)$ coincides with the intersection of all maximal right ideals of R .

Recall: M : R -module, $x \in M$, $\text{Ann}_R(x) = \{r \in R \mid rx = 0\}$ left ideal of R

$S \subseteq M$ subset, $\text{Ann}_R(S) = \bigcap_{x \in S} \text{Ann}_R(x)$ an left ideal

$N \subseteq M$ submodule $\text{Ann}_R(N) = \bigcap_{x \in N} \text{Ann}_R(x)$ an 2-sided ideal

proof: (1), let S be a simple left module of R

$$S \cong Ru \text{ for any } 0 \neq u \in S$$

$$S \cong R^R / \text{Ann}_R(v)$$

Since S is simple, $\text{Ann}_R(v)$ is a maximal left ideal.

$$J(R) \subseteq \text{Ann}_R(v) \text{ for any } v$$

hence $J(R) \subseteq \text{Ann}_R(S)$ for any simple left module S

Conversely, if $x \in \text{Ann}_R(R/I)$ then $x \in I$

so $x \in \text{Ann}_R(S)$, $\forall$ simple $S \Rightarrow x \in J(R)$.

(2) ($\Rightarrow$). Let $a \in J(R)$. Suppose $\exists x \in R$ s.t. $1 - xa$ has no left inverse. That is $R(1 - xa) \subsetneq R$

There is a maximal left ideal M of R s.t. $1 - xa \in M$

Then $xa \in M$ $1 = 1 - xa + xa \in M$. $\downarrow$

($\Leftarrow$) If $a \notin J(R)$ There is a max left ideal M of R s.t.

$a \notin M$, hence $M + Ra = R$, then $\exists b \in M, x \in R$

$b + xa = 1 \Rightarrow 1 - xa = b \in M$ has no left ideal.

(3). Let $a \in J(R)$ by (2). $\exists b \in R$. $b(1 - a) = 1$

Then $b = 1 + ba$ by (2) b has left inverse.

i.e. $\exists c \in R$, $cb = 1$

$$c = cb(1 - a) = 1 - a$$

namely $1 - a$ is a unit in R

Let I be a 2-sided ideal with $a \in I \Rightarrow 1-a$ is a unit in R

Suppose $I \not\subseteq J(R)$, i.e. $\exists$ a maximal left ideal M

such that $I \not\subseteq M$

Then $I+M=R$, $\exists a \in I, b \in M$ s.t. $a+b=1$

$b=1-a$ is a unit, contradiction.

(4). Just define $J(R)$ in right side, as (3). we find right side $\Leftrightarrow$ left side.

Definition: (1) An element $a \in R$ is called nilpotent if

$a^n = 0$ for some $n \in \mathbb{N}_+$

Note: if so, the $1-a$ is a unit. as

$$(1-a)(1+a+\dots+a^{n-1}) = 1$$

(2). A left ideal I is called nilpotent if $I^n = 0$ for some $n \in \mathbb{N}_+$

A left ideal I is called a nil left ideal if every element of I is nilpotent.

clearly any nilpotent ideal is nil

Theorem: Let I be a nil left ideal of R then $I \subseteq J(R)$

proof: let $a \in I, \forall x \in R \quad xa \in I$

xa is nilpotent $\Rightarrow 1-xa$ is a unit in R

$\Rightarrow a \in J(R)$

Theorem: Let A be Artinian. Then $J(A)$ is nilpotent

pf: Consider the chain $J(A) \supseteq J(A)^2 \supseteq \dots$

by Artinian $\exists$ n such that $J(A)^n = J(A)^{n+1}$

Assume $J(A)^n \neq 0$ denote $N := J(A)^n$

Consider the set of left ideals I of R with $NI \neq 0$

its non empty as $NN \neq 0$

Let I_0 be a minimal element of this set

$NI_0 \neq 0 \Rightarrow \exists a \in I_0$ s.t. $Na \neq 0$

$Na \subseteq I_0$ $N(Na) = Na \neq 0$

$\Rightarrow I_0 = Na$ since the minimality

$\exists b \in N$ s.t. $a = ba$

$\Rightarrow (1-b)a = 0$

recall $b \in N = J(A)^n \subseteq J(A)$, $1-b$ is a unit. $\downarrow$

Corollary: Let A be an Artinian ring. Then $J(A)$

is the largest nilpotent left ideal of A

Theorem: Let I be a 2-sided ideal of R with $I \subseteq J(R)$

Then $J(R/I) = J(R)/I$.

Theorem: A : Artinian ring. Then A is semisimple if and

only if $J(A) = 0$.

proof: ($\Leftarrow$) Let M_λ , $\lambda \in \Lambda$ be maximal left ideals of A

Consider the set of left ideals that are expressed as finite intersection of some M_λ

Let L be a minimal element, say $L = M_{\lambda_1} \cap \dots \cap M_{\lambda_r}$

If $L \neq 0$, then there exists M_λ with $L \cap M_\lambda \subsetneq L$ since $J(A) = 0$

But L is minimal, $\downarrow$ Hence $L = 0$

$\Rightarrow$ finite many M_{λ_i} , intersection will be zero

Consider $A \rightarrow A/M_{\lambda_1} \oplus \dots \oplus A/M_{\lambda_r}$

this is an A -module monomorphism

Hence A is semisimple

($\Rightarrow$)

If A is semisimple, then ${}_A A = S_1 \oplus \dots \oplus S_k$

S_i 's are simple A -module

$$J(A) \subseteq \text{Ann}_A(S_i) \Rightarrow J(A)A = 0 \Rightarrow J(A) = 0$$

Corollary: Let A be an Artin ring

Then $J(A)$ is the unique left ideal U of A with the

property that

• U is nilpotent, and

• A/U is semisimple.

proposition: R Artin ring, U : finite generated R -module.

(i) the followings are equivalent for R -submodule of U

(1) $V = \text{Rad}(U)$

(2) V is the smallest submodule of U with semisimple quotient

(3) $J(R) \cdot U = V$

(ii) the followings are equivalent for an R -submodule V of U

(1) $V = \text{Soc}(U)$

(2) V is the largest semisimple submodule of U

(3) $V = \{u \in U \mid J(R) \cdot u = 0\}$

proof: (i) (1) $\Leftrightarrow$ (2) (ii) (1) $\Leftrightarrow$ (2)

(ii) (2) $\Leftrightarrow$ (3) consider $V = \{u \in U \mid J(R) \cdot u = 0\}$

regard V as $R/J(R)$ -module.

hence V is semisimple $R/J(R)$ -module.

hence V is also semisimple R -module.

Conversely any simple submodule of U can be annihilated by $J(R)$

(i). (3) $\Rightarrow$ (2) Let $V = J(R) \cdot U$ then U/V is a $R/J(R)$ -module

hence semisimple

If V' is a submodule of U with semisimple quotient

$$J(R) \cdot (U/V') = 0 \Rightarrow J(R) \cdot U \subseteq V'$$

(2) $\Rightarrow$ (3) If $V' \subseteq U$ is a maximal submodule

then $J(R) \cdot (U/V') = 0 \Rightarrow J(R)U \subseteq V'$

So $J(R) \cdot U \subseteq \text{Rad}(U)$

Since $U/J(R) \cdot U$ is semisimple, hence $\text{Rad } U \subseteq J(R) \cdot U$

Let R be an Artin ring, consider finitely generated module U .

Define inductively: $\text{Rad}^n(U) = \text{Rad}(\text{Rad}^{n-1}(U))$

$$\text{Soc}^n(U) / \text{Soc}^{n-1}(U) = \text{Soc}(U / \text{Soc}^{n-1}(U))$$

$$\Rightarrow \text{Rad}^n(U) = J(R)^n \cdot U \quad \text{Soc}^n(U) = \{u \in U \mid J(R)^n \cdot u = 0\}$$

We can get two chains of submodule

$$\begin{array}{ll} \text{Rad } U \supseteq \text{Rad}^2 U \supseteq \dots & \text{radical series} \\ \text{Soc } U \subseteq \text{Soc}^2 U \subseteq \dots & \text{socle series} \end{array} \left\{ \begin{array}{l} R\text{-artin they} \\ \text{are finite} \\ \text{since } J(R)^n = 0 \end{array} \right.$$

Consider $\text{Rad}^{n-1} U / \text{Rad}^n U$: radical layers

$\text{Soc}^n U / \text{Soc}^{n-1} U$ = socle layers

Lemma: $\text{Rad}^n(U \oplus V) = \text{Rad}^n(U) \oplus \text{Rad}^n(V)$

$\text{Soc}^n(U \oplus V) = \text{Soc}^n(U) \oplus \text{Soc}^n(V)$

Theorem: A : Artin ring U : A -module

The radical series of U is the farthest descending chain

of submodules of U with semisimple quotient
and the socle series of U is the farthest ascending series
with semisimple quotient.

The two series terminate

and if m and n are the minimal integer s.t.

$$\text{Rad}^m U = 0 \quad \text{and} \quad \text{Soc}^n U = U \quad \text{then} \quad m =$$

proof: Suppose $\dots \subseteq U_r \subseteq U_{r-1} \subseteq \dots \subseteq U_0 = U$ series with semisimple
quotients. We show by induction that $\text{Rad}^r U \subseteq U_r$

$r=1$ is obvious

sps for $r > 0$, $\text{Rad}^{r-1} U \subseteq U_{r-1}$

$$\text{Rad}^{r-1}(U) / \text{Rad}^{r-1}(U) \cap U_r \cong \text{Rad}^{r-1}(U) + U_r / U_r \subseteq U_{r-1} / U_r$$

semisimple. $\Rightarrow \text{Rad}^{r-1}(U) \cap U_r \supseteq \text{Rad}^r(U)$

$$\Rightarrow \text{Rad}^r(U) \subseteq U_r$$

Similarly for socle series.

Since $J(A)$ nilpotent let n be the integer $J(A)^n = 0$, $J(A)^{n-1} \neq 0$

Then $\text{Rad}^n(U) = 0$ $\text{Rad}^{n-1} U \neq 0$

$$\text{Soc}^n(U) = U \quad \text{Soc}^{n-1} U \neq U$$

Let I be a 2-sided ideal of R , f idempotent of R , $f \notin I$

Then $e = f + I$ is an idempotent of R/I . We say f lifts e

Remark: Given an idempotent e of R/I , it may or may not be possible to lift it to an idempotent of R

If for every idempotent e of R/I there is an idempotent $f \in R$ such that f lifts e , we say we can lift idempotents from R/I to R

Theorem: let I be a nilpotent 2-sided ideal of R and $e \in R/I$ idempotent. Then there is an idempotent $f \in R$ lifting e

If e is primitive, so is any lifting f

proof: define idempotent $e_i \in R/I^i$ inductively

such that $e_i + I^{i-1}/I^i = e_{i-1}$ for all i , $e_1 = e$

sps e_{i-1} is an idempotent of R/I^{i-1}

Pick any element $a \in R/I^i$ with $a + I^{i-1} = e_{i-1}$ so

$a^2 - a \in I^{i-1}/I^i$ observe $(I^{i-1})^2 \subseteq I^i \Rightarrow (a^2 - a)^2 = 0$

in R/I^i

Put $e_i = 3a^2 - 2a^3$ $e_i + I^{i-1} = e_{i-1}$

$e_i^2 - e_i = 0$ in R/I^i

hence e_i is an idempotent in R/I^i

If $I^r = 0$, then $e_r = f$